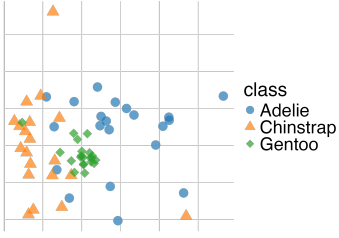
# Introduction to Machine Learning

# Random Forest Proximities




**Learning goals**

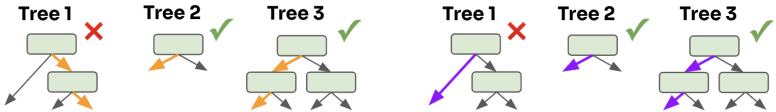
- Understand how RF can be used to define proximities of observations
- Know how proximities can be used for visualization, outlier detection and imputation

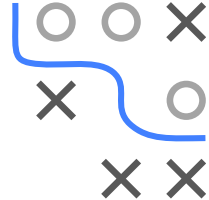
# PROXIMITIES

RFs have built-in similarity measure for pairs of observations:

| ID | Color | Form | Length |
|---|---|---|---|
| 1 | yellow | oblong | 14 |

| ID | Color | Form | Length |
|---|---|---|---|
| 2 | brown | oblong | 10 |



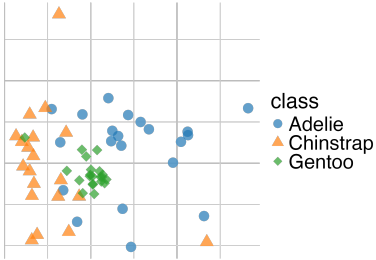



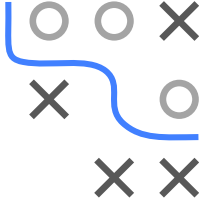
- After training, push all observations through each tree
- To calculate $\text{prox}\left(\mathbf{x}^{(i)}, \mathbf{x}^{(j)}\right)$: Percentage of how often both points are placed in **same terminal node of a tree**
- Here: $\text{prox}\left(\mathbf{x}^{(1)}, \mathbf{x}^{(2)}\right) = 2/3$
- All proximities are arranged in symmetric $n \times n$ matrix

# VISUALIZING PROXIMITIES

| obs. \ observation | 1 | 2 | ... | n |
|---|---|---|---|---|
| 1 | 1 | 0.2 | ... | 0.5 |
| 2 | 0.2 | 1 | ... | 0.8 |
| ... | ... | ... | ... | ... |
| n | 0.5 | 0.8 | ... | 1 |

MDS



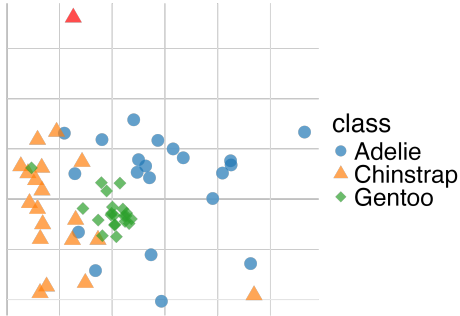



Can visualize the proximity matrix by projecting it into lower-dim. space, e.g., via multidim. scaling (might have to turn proximities into distances)

- Samples from same class usually form **identifiable clusters**
- **Offers some error-inspection**, e.g., Adelie has high within-class variance and has overlaps with other classes

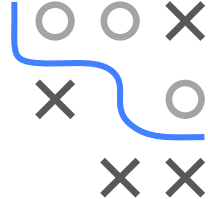
# OUTLIER DETECTION

- Can also be used to **locate outliers**
- Or mislabeled points, especially in manually labeled data sets



# IMPUTING MISSING DATA

| ID | Color | Form | Origin | Length |
|---|---|---|---|---|
| 1 | yellow | round | domestic | 14 |
| 2 | brown | oblong | imported | ??? |
| 3 | brown | oblong | imported | 19 |
| 4 | ??? | round | domestic | 14 |



1. Replace missings per feature by median (of available values)
2. Compute proximities (NB: data has changed)
3. Replace missings in $\mathbf{x}^{(i)}$ by weighted average of non-missings; weights proportional to proximities

Steps 2 and 3 are iterated a few times.

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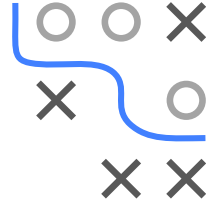

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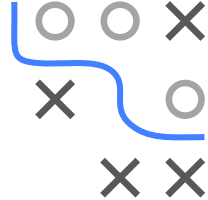
weighted average
using proximities



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